

A Construction of Deformations to General Algebras

David Bowman, Dora Puljić*, and Agata Smoktunowicz

School of Mathematics, JCMB, The King's Buildings, University of Edinburgh, EH9 3BF, Edinburgh, UK

*Correspondence to be sent to: e-mail: D.puljic@sms.ed.ac.uk

One of the questions investigated in deformation theory is to determine to which algebras can a given associative algebra be deformed. In this paper we investigate a different but related question, namely: for a given associative finite-dimensional $\mathbb{C}$ -algebra A , find algebras N , which can be deformed to A . We develop a simple method that produces associative and flat deformations to investigate this question. As an application of this method we answer a question of Michael Wemyss about deformations of contraction algebras.

1 Introduction

Michael Wemyss and Will Donovan developed a method for characterising commutative rings that appear in algebraic geometry, by using methods from noncommutative ring theory. In [6] they introduced contraction algebras that provide important insight into resolutions of noncommutative singularities and invariants of flops. For information about contraction algebras we refer reader to [6, 14]. Contraction algebras can be described in a purely algebraic way, by using generators and relations (see Section 4). They appear in many questions, which can be investigated by noncommutative ring theorists who are not familiar with advanced methods of algebraic geometry. Some of these questions are related to nilpotent rings. In 2022, Gavin Brown and Michael Wemyss described all finite dimensional contraction algebras that are local [2]. It is known that an algebra R is local if the factor algebra $R/J(R)$ is a field, where $J(R)$ is the Jacobson radical of R . Note that the Jacobson radical of a finite dimensional algebra is nilpotent. Therefore, contraction algebras that are local and finite dimensional are close to being nilpotent. Some other questions involve characterisation of contraction algebras that are not local.

Deformations of contraction algebras give insight into invariants of flops. In this context Michael Wemyss asked questions about deformations of contraction algebras that are local. Using geometric methods he was able to conjecture which rings can be obtained as deformations of these contraction algebras [13].

Since contraction algebras are noncommutative, the usual methods using derivations as in [3] cannot be applied. Another approach can be to use Hochschild cohomology and the Gerstenhaber bracket; however, this often leads to complicated calculations as seen in [11].

The geometric deformations considered in our paper that are related to Michael Wemyss' questions are described on page 29 [10]. Also, on page 7 in [12] Toda explains the existence of a geometric deformation. Here we follow some information provided by Michael Wemyss in [13]. In Theorem 4.2 [9], Hua and Toda showed that the geometric deformation related to the Atiyah flop of the

Received: July 31, 2023. Revised: February 14, 2024. Accepted: March 13, 2024

© The Author(s) 2024. Published by Oxford University Press.

This is an Open Access article distributed under the terms of the Creative Commons Attribution License (<http://creativecommons.org/licenses/by/4.0/>), which permits unrestricted reuse, distribution, and reproduction in any medium, provided the original work is properly cited.

contraction algebra Λ_{con} is semi-simple. So it suggests that the geometric deformation gives an algebraic deformation of the contraction algebra Λ_{con} . Yet we do not quite know this. Hua and Toda say in their introduction: “We are still not able to prove that our deformations of the contraction algebras are flat...”, and they conjecture that it is true. This conjecture is still open; until flatness is proven it is not known whether or not the geometric deformation produces an algebraic one. There should be many more algebraic deformations since the Hochschild cohomology of Λ_{con} is large. It is not clear what makes the “geometric one” special, and it is not known what information any other possible algebraic deformations of Λ_{con} will give. One of the motivations for our paper was to give some methods of constructing algebraic deformations of contraction algebras.

Using information from geometry, Michael Wemyss was able to conjecture that a given contraction algebra should deform flatly to a given semisimple algebra. We consider this in Section 4 for one of the contraction algebras suggested by Michael Wemyss in [13]. More precisely, we develop a method to calculate deformations of noncommutative local algebras, using which we answer the question of does the algebra

$$\frac{\mathbb{C}\langle x, y \rangle}{(xy + yx, x^3 + y^2, y^3)}$$

deform into $A = M_2(\mathbb{C}) \oplus \mathbb{C}^{\oplus 5}$? Note that our method described in Section 3 might be applied to other contraction algebras described in [2, Section 8.1].

In [11] deformations of graded Hecke algebras were constructed using novel methods. It is known that deformations of graded Hecke algebras give important information about the Hecke algebras. Our method could be also used as another method of choice for constructing deformations of graded Hecke algebras. Some other methods of constructing deformations of associative algebras were described in [7, 15]. We would like to ask:

Question 0.1. Suppose that there is an associative and flat deformation of polynomial type from a finite dimensional algebra N to a semisimple finite dimensional algebra A . Does it follow that there is a deformation from N to A obtained by this method, or by a generalisation of this method obtained by changing the function f , where f is defined as in Notation 1.7?

1 Preliminaries

Definition 1.1. Let S be a subset of a k -algebra A for a commutative unital ring k . Then S is a **generating set** if all elements of A can be written as a k -linear sum of products of elements of S using the operations in A .

Definition 1.2 ([15]). A **formal deformation** $(A_t, *)$ of a k -algebra A is an associative k -bilinear multiplication $*$ on the $k[t]$ -module $A[t]$, such that in the specification at $t = 0$, the multiplication corresponds to that on A ; this multiplication is required to be determined by a multiplication on elements of A and extended to $A[t]$ by the Cauchy product rule.

Any multiplication $*$ as in Definition 1.2 is determined by products of pairs of elements of A : for $a, b \in A$, write

$$a * b = \sum_{i \geq 0} \mu_i(a \otimes b)t^i,$$

where μ_0 denotes the usual product in A and μ_i for $i \geq 1$ are k -linear functions from $A \otimes A$ to A . **We will say that this deformation is of polynomial type if there is m such that $\mu_i = 0$ for all $i > m$.**

Definition 1.3 ([7]). Let A be a k -algebra and for $t \in \mathbb{C}$ let $\{A_t\}$ be the family of algebras arising from a deformation $*$ of A . Then $*$ is a **flat** deformation if each A_t has the same dimension.

Notation 1.4. We will denote by $\mathbb{C}\langle x, y \rangle$ the polynomial ring over $\mathbb{C}$ in two non-commuting variables, x and y .

Notation 1.5. Consider monomials of the ring $\mathbb{C}\langle x, y \rangle$. We can order the monomials using shortlex ordering by specification $x < y$ and setting 1 as the least monomial. Note that this means that if a monomial $p_i \in \mathbb{C}\langle x, y \rangle$ is a product of n_i elements from the set $\{x, y\}$, then $p_i < p_j$ whenever $n_i < n_j$. If $n_i = n_j$, then we can let $p_i = a_1 a_2 \dots a_{n_i}$ and $p_j = b_1 b_2 \dots b_{n_j}$, where $a_m, b_m \in \{x, y\}$. Then $p_i < p_j$ if and only if $a_k < b_k$, where k is the first index where a_m and b_m differ. We denote the monomials of $\mathbb{C}\langle x, y \rangle$ by $p_1, p_2 \dots$ where $p_i < p_j$ for $i < j$.

Notation 1.6. We will denote by $\mathbb{C}\langle x, y \rangle[t]$ the polynomial ring in variable t with coefficients from the polynomial ring in two non-commuting variables x and y , with coefficients from $\mathbb{C}$.

Notation 1.7. Let A be a $\mathbb{C}$ -algebra. For $a_i, b_i \in A$ we define $f : \mathbb{C}\langle x, y \rangle[t] \rightarrow A[t]$ to be a homomorphism of $\mathbb{C}$ -algebras such that

$$f : x \mapsto \sum_{i=1}^n a_i t^i, \quad f : y \mapsto \sum_{i=1}^n b_i t^i, \quad f : t \mapsto t.$$

We will denote $\ker f$ by I .

Notation 1.8. For an element r in $\mathbb{C}\langle x, y \rangle$ we will denote by $\bar{r}$ the same element as r , but with all instances of x replaced by $\sum_{i=1}^n a_i$, and all instances of y replaced by $\sum_{i=1}^n b_i$.

Notice that $\bar{\bar{p}_i} = f(p_i)(1)$, for f as in Notation 1.7.

Notation 1.9. Let R be a ring and I be an ideal of R , then elements of the factor ring R/I will be denoted as $r + I$, where $r \in R$. Notice that $r + I = s + I$ for $r, s \in R$ if and only if $r - s \in I$.

2 Algorithm

In this section we shall describe the simplest case of our proposed method. This requires the data of a $\mathbb{C}$ -algebra A and a two element generating set $\{a, b\}$. We shall state the steps of this algorithm and then provide notes regarding important components. Then some examples will be displayed.

Note that In Section 5 we show that the method described here can be generalised to finite dimensional $\mathbb{C}$ -algebra A with n generators.

Method 1. We fix a finite dimensional $\mathbb{C}$ -algebra A with two generators a, b . Then:

- 1) We consider $A[t]$, the polynomial algebra over A . We will use t as the parameter of our deformation. We define:

$$x := ta,$$

$$y := tb.$$

- 2) We calculate:

$$x^2 = t^2 a^2,$$

$$xy = t^2 ab,$$

$$yx = t^2 ba,$$

$$y^2 = t^2 b^2,$$

and continue to calculate larger products of x, y . We shall then proceed with a Diamond Lemma (See section 1 of [1].) like decomposition of large products of a, b in terms of smaller products of a, b . This will cause all elements of sufficiently large length to have a power of t as a factor. In

doing so we will obtain relations on x, y and products thereof. We terminate this process when we have enough relations to decompose any large product into only multiples of x, y and powers of t . Our relations will be given by polynomials $p_1, \dots, p_m \in \mathbb{C}\langle x, y \rangle[t]$ for some $m \in \mathbb{N}$. This terminates finitely because A finite dimensional.

3) We present the algebra:

$$\mathcal{N} := \mathbb{C}\langle x, y \rangle[t] / \langle p_1, \dots, p_m \rangle.$$

We note that sufficiently large products of x, y will obtain a factor of t . Now we evaluate t at various values in $T = [0, \infty)$. We denote by N_s the algebra that arises from $\mathcal{N}$ by evaluating t at s and in particular we write $N = N_0$. By step (2) N is local, in Section 5 we consider a more general method where N is not necessarily local. The algebra $\mathcal{N}$ and the family of algebras $\{N_t\}$ are the output of this method.

By Theorem 3.7 the family $\{N_t\}$ is a deformation of N and by Proposition 3.9 and Theorem 3.10 it holds that $A \in \{N_t\}$.

Changing the chosen generators a, b can result in different algebras N_0 that have A as an associative deformation.

In our proofs regarding the general case, in which A is an associative finite dimensional $\mathbb{C}$ -algebra generated by some elements $\sum_{i=1}^n a_i, \sum_{i=1}^n b_i$ for some $a_i, b_i \in A$, we use steps analogous to the ones described above. Point 1 corresponds to Notation 1.7. Most of section 3 (up to Theorem 3.7) corresponds to point 2, and Theorem 3.7, Proposition 3.9 and Theorem 3.10 correspond to point 3. Moreover, in section 4 we answer a question of Wemyss as an application, and in section 5 we generalise Theorem 3.7 to the case of many variables.

2.1 Notes on the method

We give some notes before the examples:

- In step (1) we are trying to capture the behaviour of the elements in A , but in a way that is controlled by t . This enables us to make use of a presentation while still having access to our deformation parameter t .
- We note that since a, b generate A some linear combination of products of a, b will be the identity. In step (2) it is important we find a relation on x, y, t and $1 \in A$.
- In step (3) we discard A entirely. However, because of the relations p_i , x and y “remember” that they came from A and that they form a generating set. This step results in something subtle: the monomials $x, y \in N$ are no longer dependent on t (and thus do not vanish when we set $t = 0$) but their products maintain their dependence on t . We have loosened our grip on x, y just enough for the deformation to take place.
- We note that our deformation will be associative since the multiplication of N_t will inherit from the associative multiplication $*$ of the algebra $\mathcal{N}$.
- This algorithm is easily generalised to algebras that require a larger generating set. Suppose a finite dimensional $\mathbb{C}$ -algebra A has n generators $\{a_1, \dots, a_n\}$. Then in step (1) we would define $x_i := ta_i$ for $1 \leq i \leq n$ and proceed as described in Method 1. The resulting family of algebras will also be a flat and associative deformation of A .

2.2 Examples

Here we shall work through some examples. We shall start with a simple example where $A = \mathbb{C} \oplus \mathbb{C}$ and then give a more complicated one where $A = M_2(\mathbb{C})$, the algebra of 2×2 matrices over $\mathbb{C}$. Practically it can be useful to fix a basis for the algebra A as this helps find the decompositions in step (2).

Example 2.1. We consider $A = \mathbb{C} \oplus \mathbb{C}$ and fix $a = (i, 0)$, $b = (0, 1)$. Clearly $\mathbb{C}$ -linear combinations of a and b span A so our basis is just $\{a, b\}$. We write $1 := (1, 1)$. Now we consider $A[t]$ and define:

$$x := ta = (ti, 0),$$

$$y := tb = (0, t).$$

We calculate:

$$\begin{aligned}x^2 &= t^2 a^2 = (-t^2, 0) = tix, \\xy &= t^2 ab = (0, 0) = 0, \\yx &= t^2 ab = (0, 0) = 0, \\y^2 &= t^2 b^2 = (0, t^2) = ty, \\t\mathbb{1} &= -ix + y.\end{aligned}$$

Thus we have relations:

$$\begin{aligned}p_1 &= x^2 - tix = 0, \\p_2 &= xy = 0, \\p_3 &= yx = 0, \\p_4 &= y^2 - ty = 0, \\p_5 &= t\mathbb{1} - (-ix + y).\end{aligned}$$

Since $xy = yx = 0$ and squares of x, y can be reduced, we have enough relations to decompose any large product into only multiples of x, y and powers of t . Thus we present:

$$N := \mathbb{C}\langle x, y \rangle[t] / \langle p_1, p_2, p_3, p_4, p_5 \rangle.$$

We observe:

$$N_0 = \mathbb{C}\langle x, y \rangle / \langle x^2, xy, yx, y^2, -ix + y \rangle,$$

and

$$N_1 = \mathbb{C}\langle x, y \rangle / \langle x^2 - ix, xy, yx, y^2 - y, 1 + ix - y \rangle.$$

We see that x, y behave exactly as $a, b \in A$ and by Proposition 3.8. It is easily checked that the isomorphism is given by:

$$\begin{aligned}\phi : N_1 &\rightarrow A \\1 &\mapsto \mathbb{1} \\x &\mapsto (i, 0) \\y &\mapsto (0, 1).\end{aligned}$$

Thus we have produced an algebra N_0 that has $A = \mathbb{C} \oplus \mathbb{C}$ as a deformation.

Example 2.2. We consider $A = M_2(\mathbb{C})$ and fix:

$$a = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad b = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

We denote by $\mathbb{1}$ the multiplicative unit in A . Next we compute:

$$\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} = \frac{1}{2}(a + b^2) = e_1,$$

$$\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = \frac{1}{2}(ab + b) = e_2,$$

$$\begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} = \frac{1}{2}(b - ab) = e_3,$$

$$\begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} = \frac{1}{2}(b^2 - a) = e_4$$

and thus a, b generate A as an algebra. Now we consider $A[t]$ and define:

$$x := ta,$$

$$y := tb.$$

We compute:

$$x^2 = t^2 \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = t^2 \mathbb{1},$$

$$xy = t^2 \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = t^2(e_2 - e_3),$$

$$yx = t^2 \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = t^2(e_3 - e_2),$$

$$y^2 = t^2 \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = t^2 \mathbb{1}.$$

We have obtained the relations:

$$p_1 = x^2 - t^2 \mathbb{1},$$

$$p_2 = x^2 - y^2 = 0,$$

$$p_3 = xy + yx = 0,$$

$$p_4 = y^2 - t^2 \mathbb{1}.$$

Now we consider larger products to produce more relations:

$$x^3 = t^3 \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = t^2 x,$$

$$y^3 = t^3 \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = t^2 y.$$

Hence we have also obtained:

$$p_5 = x^3 - t^2 x = 0,$$

$$p_6 = y^3 - t^2 y = 0.$$

These allow us to derive the following:

$$x^2 y = y^3 = y x^2 = y^3 = t^2 y,$$

$$y^2 x = x^3 = x y^2 = x^3 = t^2 x,$$

which are enough to reduce an arbitrary product of x, y . Thus we present:

$$N := \mathbb{C}\langle x, y \rangle[t] / \langle p_1, p_2, p_3, p_4, p_5, p_6 \rangle.$$

We observe:

$$N_0 = \mathbb{C}\langle x, y \rangle / \langle x^2, y^2, xy + yx \rangle,$$

and

$$N_1 = \mathbb{C}\langle x, y \rangle / \langle x^2 - 1, y^2 - 1, xy + yx, x^3 - x, y^3 - y \rangle.$$

By Proposition 3.8 we have $N_1 \cong A$. It is easily checked that the isomorphism is given by:

$$\begin{aligned} \phi : N_1 &\rightarrow A \\ 1 &\mapsto \mathbb{1} \\ x &\mapsto \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \\ y &\mapsto \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}. \end{aligned}$$

Thus we have produced an algebra N_0 that has $A = M_2(\mathbb{C})$ as a deformation.

3 Deformations to A

We now consider the general case in which A is an associative finite dimensional $\mathbb{C}$ -algebra generated by some elements $\sum_{i=1}^n a_i, \sum_{i=1}^n b_i$ for some $a_i, b_i \in A$.

3.1 Specification at $t = 0$

In this section we define a multiplication on $\mathbb{C}\langle x, y \rangle[t]$, which gives the algebra N at specification $t = 0$.

Proposition 3.1. Let $\mathbb{C}[t]$ denote the polynomial ring in variable t . Recall that f is as per Notation 1.7. Further, assume that for every $r \in A$ there exists $i = i(r)$ such that $rt^i \in \text{im} f$. Then there exist elements $q_1, q_2, \dots, q_n \in \mathbb{C}\langle x, y \rangle[t]$ such that for every $r \in \mathbb{C}\langle x, y \rangle[t]$ we have

$$f(r) \in \sum_{i=1}^n \mathbb{C}[t]f(q_i). \quad (1)$$

Moreover $n = \dim A$.

Suppose $\sum_{i=1}^n \alpha_i q_i \in I$ for some $\alpha_i \in \mathbb{C}[t]$. Then $f(\sum_{i=1}^n \alpha_i q_i) = 0$ and consequently $\alpha_i = 0$ for every $i \in \{1, 2, \dots, n\}$.

Moreover, all elements of I are $\mathbb{C}[t]$ -linear combinations of elements

$$p_k - \sum_{i=1}^n (\zeta_{i,k} q_i + t \xi_{i,k}(t) q_i)$$

for some $k \in \mathbb{N}$, $\zeta_{i,k} \in \mathbb{C}$, $\xi_{i,k}(t) \in \mathbb{C}[t]$.

Proof. We let $\{e_1, \dots, e_n\}$ be a basis of A . Let k_1 be the smallest possible integer such that for some integer j_1 we have

$$e_{j_1} t^{k_1} + \sum_{i \neq j_1} \alpha_i(t) e_i \in \text{im} f$$

for some $\alpha_i(t) \in \mathbb{C}[t]$. Let k_2 be the smallest possible integer such that for some integer $j_2 \neq j_1$ we have

$$e_{j_2} t^{k_2} + \sum_{i \neq j_1, j_2} \alpha'_i(t) e_i \in \text{im} f$$

for some $\alpha'_i(t) \in \mathbb{C}[t]$. Continuing in this way, we define a basis $\{q_1, \dots, q_n\}$ such that

$$f(q_m) = e_{j_m} t^{k_m} + \sum_{i \neq j_1, \dots, j_m} \alpha_i(t) e_i.$$

Now notice that $f(q_1), f(q_2), \dots, f(q_n)$ are linearly independent over $\mathbb{C}$ as $e_1, e_2, \dots, e_n$ are linearly independent over $\mathbb{C}$.

We now prove that $f(q_1) + \text{im} f, \dots, f(q_n) + \text{im} f$ span the $\mathbb{C}$ -algebra $\frac{\text{im} f}{\text{im} f}$ over $\mathbb{C}$, and hence they are a basis for $\frac{\text{im} f}{\text{im} f}$. Let $z_1 \in \text{im} f$ and write

$$z_1 = \sum_{i=1}^n r_i(t) e_{j_i} t^{s_i}$$

for some $r_i \in \mathbb{C}[t] - t\mathbb{C}[t]$ and some $j_i, s_i \in \mathbb{N}$.

Notice that for the smallest i such that $r_i(t) \neq 0$, $s_i \geq k_i$ for k_i as above. Let

$$z_2 = z_1 - f(q_1) t^{s_1 - k_1} r_1(t).$$

Note that we have

$$z_2 \in \sum_{i=2}^n \mathbb{C}[t] e_{j_i}$$

so we can write

$$z_2 = \sum_{i=2}^n r_i^2(t) e_{j_i} t^{s_i^2}$$

for some $r_i^2(t) \in \mathbb{C}[t] - t\mathbb{C}[t]$ and $s_i^2 \in \mathbb{N}$. Notice that $q_2 \in \text{im} f$. Now let

$$z_3 = z_2 - f(q_2) t^{s_2^2 - k_2} r_2^2(t).$$

Continuing in this way we eventually arrive at $q_{n+1} = 0$. Hence by summing the following equations

$$0 = z_n - f(q_n) t^{s_n^n - k_n} r_n^n(t),$$

$$\vdots$$

$$z_3 = z_2 - f(q_2) t^{s_2^2 - k_2} r_2^2(t)$$

and

$$z_2 = z_1 - f(q_1) t^{s_1 - k_1} r_1(t)$$

we arrive at

$$z_1 = \sum_{l=1}^n f(q_l) t^{s'_l - k_l} r'_l(t) \in \sum_{l=1}^n \mathbb{C}[t] f(q_l)$$

for some $s'_l \in \mathbb{N}$ as required. This shows that $\text{im} f \subseteq \sum_{i=1}^n \mathbb{C}[t] q_i$.

Now suppose $\sum_{i=1}^n \alpha_i q_i \in I$. Recall that $I = \ker f$, so

$$0 = f\left(\sum_{i=1}^n \alpha_i q_i\right) = \sum_{i=1}^n \alpha_i f(q_i).$$

Recall that

$$f(q_m) = e_{j_m} t^{k_m} + \sum_{i \neq j_1, \dots, j_m} \alpha_i(t) e_i$$

for each $m \in \mathbb{N}$. Now suppose that not all α_i equal 0. Let m' be the maximal index such that $\alpha_{m'} \neq 0$. Then

$$0 = f\left(\sum_{i=1}^n \alpha_i q_i\right) = \sum_{i=1}^{m'} \alpha_i f(q_i) = e_{j_{m'}} t^{k_{m'}} \alpha_{m'} + e,$$

where $e \in \sum_{i=m'+1}^n \mathbb{C}[t] e_{j_i}$. Recall that $e_{j_1}, e_{j_2}, \dots, e_{j_n}$ is the basis of A , so it is also the basis of $A[t]$ as a module over the commutative ring $\mathbb{C}[t]$. Therefore $e_{j_{m'}} t^{k_{m'}} \alpha_{m'} + e \neq 0$, which is a contradiction.

We now show that all elements of I are $\mathbb{C}[t]$ -linear combinations of elements

$$p_k - \sum_{i=1}^n (\zeta_{i,k} q_i + t \xi_{i,k}(t) q_i)$$

for some $k \in \mathbb{N}$, $\zeta_{i,k} \in \mathbb{C}$, $\xi_{i,k}(t) \in \mathbb{C}[t]$. Here we let $t_k := \sum_{i=1}^n (\zeta_{i,k} q_i + t \xi_{i,k}(t) q_i)$. Recall that by Notation 1.5 $\mathbb{C}(x, y) = \mathbb{C}p_1 + \mathbb{C}p_2 + \dots$. Let $r \in I$. Then $r = \sum_{i=1}^t \beta_i p_i$ for some $\beta_i \in \mathbb{C}$. Observe that $p_i - t_i \in I$, so

$$\sum_{i=1}^t \beta_i p_i = r - \sum_{i=1}^t \beta_i (p_i - t_i) \in I.$$

Note that

$$\sum_{i=1}^t \beta_i p_i \in \sum_{i=1}^n q_i \mathbb{C}[t] = \sum_{i=1}^n q_i \alpha_i$$

for some $\alpha_i \in \mathbb{C}[t]$. It follows that $\sum_{i=1}^n q_i \alpha_i \in I$. Recall that by an earlier result from this proof it follows that $\alpha_i = 0$, as required. ■

Remark. Notice that for $k \in \mathbb{N}$ and $\alpha_j \in \mathbb{C}[t]$

$$f(p_k) - \sum_{j=1}^n \alpha_j f(q_j) = 0$$

if and only if

$$p_k - \sum_{j=1}^n \alpha_j q_j \in I.$$

Proposition 3.2. Let p_k be as in Notation 1.5. Recall that I is as per Notation 1.7. For any $k, m \in \{1, 2, \dots, n\}$ there exist $\zeta_{i,k} \in \mathbb{C}$, $\xi_{i,k}(t) \in \mathbb{C}[t]$ such that

$$p_k - \sum_{i=1}^n (\zeta_{i,k} q_i + t \xi_{i,k}(t) q_i) \in I.$$

Moreover, there exist $\zeta_{i,k,m} \in \mathbb{C}$, $\xi_{i,k,m}(t) \in \mathbb{C}[t]$ such that

$$q_k q_m - \sum_{i=1}^n (\zeta_{i,k,m} q_i + t \xi_{i,k,m}(t) q_i) \in I.$$

Proof. For each element $p_k \in \mathbb{C}\langle x, y \rangle[t]$ we have

$$p_k \in \sum_{i=1}^n \mathbb{C}[t]q_i + I.$$

This follows by taking f^{-1} of equation (1) for $r = p_k$. Hence, for each $k \in \mathbb{N}$ there exist $\sigma_{i,k} \in \mathbb{C}[t]$ such that

$$p_k - \sum_{i=1}^n \sigma_{i,k} q_i \in I,$$

where $I = \ker f$. Notice that we can write $\sigma_{i,k}(t) = \zeta_{i,k} + t\xi_{i,k}(t)$ for some $\zeta_{i,k} \in \mathbb{C}$, $\xi_{i,k}(t) \in \mathbb{C}[t]$, as to separate the terms dependent on t . The above then becomes

$$p_k - \sum_{i=1}^n (\zeta_{i,k} q_i + t\xi_{i,k}(t)q_i) \in I.$$

Similarly, the existence of $\zeta_{i,k}, \xi_{i,k}$ follows by applying f^{-1} to equation (1) for $r = q_k q_m$. ■

Denote

$$q_i = \bar{q}_i + \tilde{q}_i,$$

where $\bar{q}_i \in \mathbb{C}\langle x, y \rangle$ and $\tilde{q}_i \in t\mathbb{C}\langle x, y \rangle[t]$. Then,

$$p_k - \sum_{i=1}^n \zeta_{i,k} \bar{q}_i \in I + t\mathbb{C}\langle x, y \rangle[t].$$

Notation 3.3. We denote by J the set consisting of $\mathbb{C}$ -linear combinations of elements

$$p_k - \sum_{i=1}^n \zeta_{i,k} \bar{q}_i \in \mathbb{C}\langle x, y \rangle, \quad (2)$$

where $\zeta_{i,k}$ are as in Proposition 3.2. Let $\langle J \rangle$ be the ideal of $\mathbb{C}\langle x, y \rangle$ generated by elements from J .

Further, let N be the quotient algebra $\mathbb{C}\langle x, y \rangle / \langle J \rangle$. Notice that by Proposition 3.2,

$$\bar{q}_k \bar{q}_m - \sum_{i=1}^n \zeta_{i,k,m} \bar{q}_i \in \langle J \rangle.$$

Lemma 3.4. Recall that I is as per Notation 1.7. We have $e \in J$ if and only if $e + e' \in I$ for some $e' \in t\mathbb{C}\langle x, y \rangle[t]$. In particular $J \subseteq I + t\mathbb{C}\langle x, y \rangle[t]$, and hence $\langle J \rangle \subseteq I + t\mathbb{C}\langle x, y \rangle[t]$.

Proof. If $e \in J$, then by Notation 3.3 $e = \sum_i \beta_i (p_k - l_k)$ where $l_k = \sum_{i=1}^n \xi_{i,k} \bar{q}_i$ and $\beta_i \in \mathbb{C}$. Then let

$$\bar{l}_k = \sum_i \zeta_{i,k} \bar{q}_i + \sum_i \xi_{i,k} q_i.$$

Then $e' = \sum_i \beta_i \bar{l}_i$ is such that $e + e' \in I$ by Proposition 3.2.

Suppose, on the other hand, that $e \in \mathbb{C}\langle x, y \rangle$ is such that $e + e' \in I$ for some $e' \in \mathbb{C}\langle x, y \rangle[t]$. Then by Proposition 3.1 we get that

$$e + e' \in \sum_{i=1}^n \mathbb{C}[t](p_i - t_i),$$

where $t_i = \sum_j (\zeta_{ik} q_j + t \xi_{ik}(t) q_j)$. So

$$e + e' = \sum_i \alpha_i (p_i - t_i)$$

for some $\alpha_i \in \mathbb{C}[t]$. Let $\alpha_i = r_i + \bar{r}_i$ where $r_i \in \mathbb{C}$ and $\bar{r}_i \in \mathbb{C}[t]t$. Denote $l_k = \sum_{i=1}^n \zeta_{ik} q_i$. By comparing terms containing t it follows that

$$e = \sum_i r_i (p_i - l_i).$$

So $e \in J$, as required.

Notice that if $e \in J$, then $e + e' \in I$, where $e' \in t\mathbb{C}\langle x, y \rangle[t]$ by a previous result of this proof. Therefore

$$e \in I - e' \in I + t\mathbb{C}\langle x, y \rangle[t].$$

Note that e is an arbitrary element from J so

$$J \subseteq I + t\mathbb{C}\langle x, y \rangle[t].$$

Note that $\langle J \rangle$ is an ideal generated by J , so it is the smallest ideal in $\mathbb{C}\langle x, y \rangle[t]$, which contains J . Since $I + t\mathbb{C}\langle x, y \rangle[t]$ is an ideal in $\mathbb{C}\langle x, y \rangle[t]$ and it contains J , it follows that

$$\langle J \rangle \subseteq \mathbb{C}\langle x, y \rangle[t] + I. \quad \blacksquare$$

Proposition 3.5. The dimension of N equals n . Moreover, elements $\bar{q}_k + J \in N$ are a basis of N as a $\mathbb{C}$ -vector space.

Proof. Recall that by Proposition 3.2

$$\mathbb{C}\langle x, y \rangle \subseteq \sum_{k=1}^n \mathbb{C}[t]q_k + I.$$

Consequently, expression (2) implies that

$$\mathbb{C}\langle x, y \rangle \subseteq \sum_{k=1}^n \mathbb{C}\bar{q}_k + \langle J \rangle,$$

and hence $\{\bar{q}_k + \langle J \rangle \mid k \in \{1, 2, \dots, n\}\}$ span N as a $\mathbb{C}$ -vector space. Therefore, the dimension of N does not exceed n .

We now show that the elements $\bar{q}_k + \langle J \rangle \in N$ for $k \in \{1, 2, \dots, n\}$ are linearly independent over $\mathbb{C}$. Suppose on the contrary that we have

$$\sum_{i=1}^n \xi_i \bar{q}_i \in \langle J \rangle$$

for some $\xi_i \in \mathbb{C}$. By Corollary 3.4 and Proposition 3.1 we have

$$\langle J \rangle \subseteq I + t\mathbb{C}\langle x, y \rangle[t] \subseteq I + t \sum_{i=1}^n \mathbb{C}[t]q_i.$$

Recall that

$$q_i - \bar{q}_i \in t\mathbb{C}\langle x, y \rangle[t] \subseteq t\left(\sum_{i=1}^n \mathbb{C}[t]q_i + I\right).$$

Therefore,

$$\sum_{i=1}^n \xi_i \bar{q}_i \in \langle J \rangle$$

implies

$$\sum_{i=1}^n \xi_i q_i \in \langle J \rangle + t\left(\sum_{i=1}^n \mathbb{C}[t]q_i + I\right) \subseteq I + t\sum_{i=1}^n \mathbb{C}[t]q_i.$$

It follows that there is $e' \in t\sum_{i=1}^n \mathbb{C}[t]q_i$ such that $\sum_{i=1}^n \xi_i q_i - e' \in I$, contradicting Proposition 3.1. ■

Observe now that the following holds:

Proposition 3.6. We can present N as an $\mathbb{C}$ -algebra with generators $d_1, \dots, d_n$, which span N as a $\mathbb{C}$ -vector space, subject to relations

$$d_k d_m = \sum_{i=1}^n \zeta_{i,k,m} d_i$$

where $\zeta_{i,k,m}$ are as in Proposition 3.2.

Proof. We let $\mathbb{C}\langle \bar{q}_1, \dots, \bar{q}_n \rangle$ be the subalgebra of $\mathbb{C}\langle x, y \rangle$ generated by elements $\bar{q}_1, \dots, \bar{q}_n \in \mathbb{C}\langle x, y \rangle$. Similarly, let $\mathbb{C}\langle d_1, \dots, d_n \rangle$ be the free $\mathbb{C}$ -algebra generated by elements $d_1, \dots, d_n$. Let $\zeta : \mathbb{C}\langle d_1, \dots, d_n \rangle \rightarrow \mathbb{C}\langle x, y \rangle / \langle J \rangle$ be defined as $\zeta(d_i) = \bar{q}_i + \langle J \rangle$.

We now show that $x + \langle J \rangle, y + \langle J \rangle \in \text{im} \zeta$. Notice that since $f(x) \in \text{im} f$ we have

$$f(x) \in \sum_{i=0}^{\infty} \mathbb{C}[t]f(q_i).$$

Hence we have $f(x) = f(x')$ for some $x' \in \sum_{i=0}^{\infty} \mathbb{C}[t]q_i$. Then $x - x' \in I$, and so $x - x'' \in J$ for some $x'' \in \sum_{i=0}^{\infty} \mathbb{C}\bar{q}_i$. Then $x'' + \langle J \rangle \in \text{im} \zeta$, and so $x + \langle J \rangle \in \text{im} \zeta$. An analogous argument shows $y + \langle J \rangle \in \text{im} \zeta$.

Let

$$J' = \ker(\zeta) \subseteq \mathbb{C}\langle d_1, \dots, d_n \rangle.$$

Then by the First Isomorphism Theorem for rings $\mathbb{C}\langle d_1, \dots, d_n \rangle / J'$ is isomorphic to $\text{im}(\zeta) = \mathbb{C}\langle x, y \rangle / \langle J \rangle = N$.

Let J'' be the ideal of $\mathbb{C}\langle d_1, \dots, d_n \rangle$ generated by elements

$$d_k d_m - \sum_{i=1}^n \zeta_{i,k,m} d_i$$

where $\zeta_{i,k,m}$ are as in Proposition 3.2. Observe that

$$\zeta\left(d_k d_m - \sum_{i=1}^n \zeta_{i,k,m} d_i\right) = \bar{q}_k \bar{q}_m - \sum_{i=1}^n \zeta_{i,k,m} \bar{q}_i + \langle J \rangle = 0 + \langle J \rangle.$$

Therefore $J'' \subseteq J'$. It follows that the dimension of $\mathbb{C}\langle d_1, \dots, d_n \rangle / J'$ does not exceed the dimension of $\mathbb{C}\langle d_1, \dots, d_n \rangle / J''$.

We now show that $J' = J''$. Notice that $\mathbb{C}\langle d_1, \dots, d_n \rangle / J''$ is at most n -dimensional, since every element can be presented as a linear combination of elements $d_i + J''$ for $i = 1, 2, \dots, n$. On the

other hand $\mathbb{C}\langle d_1, \dots, d_n \rangle/J'$ is isomorphic to $\mathbb{C}\langle x, y \rangle/I$, and hence is n -dimensional (by Proposition 3.5). Hence by comparing dimensions we get that the dimension of $\mathbb{C}\langle d_1, \dots, d_n \rangle/J''$ does not exceed the dimension of $\mathbb{C}\langle d_1, \dots, d_n \rangle/J'$. Previously we showed that $\mathbb{C}\langle d_1, \dots, d_n \rangle/J'$ does not exceed the dimension of $\mathbb{C}\langle d_1, \dots, d_n \rangle/J''$. It follows that $J' = J''$, as required. ■

We now define a multiplication on N , which gives a formal deformation to the algebra A .

Theorem 3.7. Let $d_1, \dots, d_n$ be free generators of the $\mathbb{C}$ -algebra $\mathbb{C}\langle d_1, \dots, d_n \rangle$ and suppose $\zeta_{i,k,m} \in \mathbb{C}$, $\xi_{i,k,m}(t) \in \mathbb{C}[t]$ are as in Proposition 3.2. Then the multiplication rule

$$d_k *_t d_m = \sum_{i=1}^n (\zeta_{i,k,m} d_i + t \xi_{i,k,m}(t) d_i).$$

gives a formal deformation such that $*_0$ gives the multiplication on N .

Proof. Recall relations from Proposition 3.2 that for any $k, m \in \{1, 2, \dots, n\}$ there exist $\zeta_{i,k,m} \in \mathbb{C}$, $\xi_{i,k,m}(t) \in \mathbb{C}[t]$ such that

$$q_k q_m - \sum_{i=1}^n (\zeta_{i,k,m} q_i + t \xi_{i,k,m}(t) q_i) \in I. \quad (3)$$

We introduce notation $[q_i] := q_i + I$ for elements of $\mathbb{C}\langle x, y \rangle/I$. Hence we obtain the following relations corresponding to (3), which give the multiplicative table on $\mathbb{C}\langle x, y \rangle/I$. We have

$$[q_k][q_m] - \sum_{i=1}^n (\zeta_{i,k,m} [q_i] + t \xi_{i,k,m}(t) [q_i]).$$

Notice that these relations give a multiplication

$$d_k *_t d_m = \sum_{i=1}^n (\zeta_{i,k,m} d_i + t \xi_{i,k,m}(t) d_i) \quad (4)$$

for $d_i \in \mathbb{C}\langle x, y \rangle$. Now recall that by Proposition 3.6 N is given by generators $d_1, \dots, d_n$ subject to relations $d_k d_m - \sum_{i=1}^n \zeta_{i,k,m} d_i$. Therefore, $*_0$ is the multiplication on N . ■

3.2 Specification at $t = 1$

In this section we define a multiplication on $\mathbb{C}\langle x, y \rangle[t]$, which gives a deformation of N , such that at specification $t = 1$ we get an algebra isomorphic to A .

Proposition 3.8. Recall that f is as per Notation 1.7. Let $\xi : \mathbb{C}\langle x, y \rangle[t] \rightarrow A$ be a homomorphism of $\mathbb{C}$ -algebras such that

$$\xi(x) = \sum_{i=1}^n a_i, \quad \xi(y) = \sum_{i=1}^n b_i, \quad \xi(t) = 1.$$

Further, assume that for every $r \in A$ there exists $i = i(r)$ such that $rt^i \in \text{im} f$. Then $\ker \xi = \langle I, t - 1 \rangle$.

Proof. We let $e = \sum_i \alpha_i p_i t^{\beta_i} \in \ker \xi$ where p_i are monomials in $\mathbb{C}\langle x, y \rangle$, $\alpha_i \in \mathbb{C}$ and $\beta_i \in \mathbb{N}$. We will show there exists $\gamma_i \in \mathbb{N}$ such that $\hat{e} := \sum_i \alpha_i p_i t^{\gamma_i} \in I$, so that $e \in I + \langle t - 1 \rangle$ as $e - \hat{e} \in \langle t - 1 \rangle$. This follows since

$$e - \hat{e} = \sum_i \alpha_i p_i t^{\beta_i} - \sum_i \alpha_i p_i t^{\gamma_i} = \sum_i \alpha_i p_i (t^{\beta_i} - t^{\gamma_i}) = \sum_i \alpha_i p_i t^m (t^n - 1) \in \langle t - 1 \rangle$$

for some $m, n \in \mathbb{N}$.

Recall Notation 1.8 and note that $\xi(e) = \sum_i \alpha_i \bar{p}_i = 0$. Notice that by assumption there exist $j_i \in \mathbb{N}$ such that $\bar{p}_i t^{j_i} \in \text{imf}$. We let $f(c_i) = \bar{p}_i t^{j_i}$ for some $c_i \in \mathbb{C}\langle x, y \rangle[t]$. Hence for large enough $k \in \mathbb{N}$ we have

$$f\left(\sum_i \alpha_i c_i t^{k-j_i}\right) = \sum_i \alpha_i \bar{p}_i t^k = 0. \quad \blacksquare$$

Proposition 3.9. Let notation be as in Proposition 3.8, and suppose that $\sum_{i=0}^n a_i$ and $\sum_{i=0}^n b_i$ generate A as a $\mathbb{C}$ -algebra. We have

$$\frac{\mathbb{C}\langle x, y \rangle[t]/I}{\langle t-1+I \rangle} \cong A.$$

Proof. Note that the ideal $\langle t-1+I \rangle$ of $\mathbb{C}\langle x, y \rangle[t]/I$ equals the set

$$\frac{\mathbb{C}\langle x, y \rangle[t](t-1)+I}{I} = \frac{\langle I, t-1 \rangle}{I} = \{g(t-1)+I \mid g \in \mathbb{C}\langle x, y \rangle[t]\}.$$

By the Third Isomorphism Theorem we have

$$\frac{\mathbb{C}\langle x, y \rangle[t]/I}{\langle I, t-1 \rangle/I} \cong \frac{\mathbb{C}\langle x, y \rangle[t]}{\langle I, t-1 \rangle}.$$

By Proposition 3.8 we have $\langle I, t-1 \rangle = \ker \xi$ and as ξ is onto,

$$\frac{\mathbb{C}\langle x, y \rangle[t]}{\ker \xi} \cong A$$

by the First Isomorphism Theorem. \blacksquare

We now define a multiplication on $\mathbb{C}\langle x, y \rangle[t]/\langle t-1, I \rangle$, which gives a deformation to A at $t=1$.

Theorem 3.10. Let notation be as in Proposition 3.8, and suppose that $\sum_{i=0}^n a_i$ and $\sum_{i=0}^n b_i$ generate A as a $\mathbb{C}$ -algebra. Let $d_1, \dots, d_n$ be free generators of the $\mathbb{C}$ -algebra $\mathbb{C}\langle d_1, \dots, d_n \rangle$ and suppose $\zeta_{i,k,m} \in \mathbb{C}$, $\xi_{i,k,m}(t) \in \mathbb{C}[t]$ are as in Proposition 3.2. Then the multiplication rule

$$d_k *_t d_m = \sum_{i=1}^n (\zeta_{i,k,m} d_i + t \xi_{i,k,m}(t) d_i).$$

gives a formal deformation such that $*_1$ gives the multiplication on an algebra generated by $d_1, \dots, d_n$ isomorphic to $\mathbb{C}\langle x, y \rangle[t]/\langle I, t-1 \rangle$.

Proof. We show that the algebra $\mathbb{C}\langle x, y \rangle[t]/\langle I, t-1 \rangle$ is isomorphic to the algebra $\mathbb{C}\langle d_1, \dots, d_n \rangle/I'$ where I' is the ideal of $\mathbb{C}\langle d_1, \dots, d_n \rangle$ generated by elements

$$d_k *_1 d_m - \sum_{i=1}^n (\zeta_{i,k,m} d_i + \xi_{i,k,m}(1) d_i)$$

for $\zeta_{i,k,m} \in \mathbb{C}$, $\xi_{i,k,m}(t) \in \mathbb{C}[t]$ as in Proposition 3.2.

We let $\delta : \mathbb{C}\langle d_1, \dots, d_n \rangle \rightarrow \mathbb{C}\langle x, y \rangle[t]/\langle t-1 \rangle$ be such that

$$\delta(d_i) = q_i + \langle t-1 \rangle,$$

for $i \in \{1, 2, \dots, n\}$. Observe that $I' \subseteq \ker(\delta)$ since

$$\begin{aligned} \delta \left(d_k * d_m - \sum_{i=1}^n (\xi_{i,k,m} d_i + \xi_{i,k,m}(1) d_i) \right) &= q_k * q_m - \sum_{i=1}^n (\xi_{i,k,m} q_i + \xi_{i,k,m}(1) q_i) \\ &\quad + \langle t-1 \rangle \\ &= \langle t-1 \rangle \end{aligned}$$

since $q_k * q_m - \sum_{i=1}^n (\xi_{i,k,m} q_i + \xi_{i,k,m}(1) q_i) \in \langle t-1 \rangle$.

Therefore the dimension of $\mathbb{C}\langle d_1, \dots, d_n \rangle / \ker(\delta)$ does not exceed the dimension of $\mathbb{C}\langle d_1, \dots, d_n \rangle / I'$. Notice that $\mathbb{C}\langle d_1, \dots, d_n \rangle / I'$ is spanned by elements $d_i + I$ as a vector space, and hence has dimension at most n . On the other hand, by the First Isomorphism Theorem for rings:

$$\mathbb{C}\langle d_1, \dots, d_n \rangle / \ker(\delta) \cong \text{im}(\delta) = \mathbb{C}\langle x, y \rangle[t]/\langle t-1 \rangle, \quad (5)$$

which in turn is isomorphic to A by Proposition 3.8, and hence has dimension n . It follows that $I' = \ker(\delta)$ and hence $\mathbb{C}\langle d_1, \dots, d_n \rangle / \ker(\delta)$ is isomorphic to $\mathbb{C}\langle d_1, \dots, d_n \rangle / I'$. Where the equality in Equation 5 is true by the proof of Proposition 3.6. ■

Remark 3.11. *Deformations at $t \neq 0$* The results from this section, hold with analogous proofs for a specialisation at other $t \neq 0$. Let $z \in \mathbb{C}$ and $z \neq 0$, the the deformation at $t = z$ is also isomorphic to A , provided that $\sum_{i=0}^n a_i z^i$ and $\sum_{i=0}^n b_i z^i$ generate A as a $\mathbb{C}$ -algebra.

Remark 3.12. Since the dimension of all the algebras in $\{N_t\}_{t \in [0,1]}$ remains $n = \dim(A)$ the deformation arising from $*$ is flat.

4 Application

In this section we prove a conjecture of M. Wemyss [13] using the method in section 3. The conjecture concerns a contraction algebra obtained from the potential $\frac{1}{4}x^4 + x \cdot y^2$. Before giving the conjecture, we give background information on contraction algebras.

We will now recall the definition of the Jacobi algebra and recall how is it connected to the question of Michael Wemyss, which we considered in this section.

Definition 4.1. Let $\mathbb{C}\langle x, y \rangle$ denote the $\mathbb{C}$ -algebra of power series in noncommuting variables x, y (the case of arbitrary number of variables is considered in [2], page 1). Let

$$f = m_1 m_2 \dots m_k$$

for some k , where $m_1, m_2, \dots, m_k \in \{x, y\}$. Define

$$f_1 = m_1 \dots m_k, \quad f_2 = m_2 \dots m_k m_1, \quad \dots \quad f_k = m_k m_1 \dots m_{k-1}.$$

We can remove the first element m_i in each f_i and denote the obtained element as h_i , so

$$h_1 = m_2 \dots m_k, \quad h_2 = m_3 \dots m_k m_1, \quad \dots, \quad h_k = m_1 \dots m_{k-1}.$$

It is possible to cyclically differentiate element f , with the partial derivatives $\frac{\delta f}{\delta x}$ and $\frac{\delta f}{\delta y}$ as follows:

$$\frac{\delta f}{\delta x} = \sum_{i: 1 \leq i \leq k, m_i = x} h_i, \quad \frac{\delta f}{\delta y} = \sum_{i: 1 \leq i \leq k, m_i = y} h_i.$$

By extending this by linearity we can cyclically differentiate any element $f \in \mathbb{C}\langle x, y \rangle$, and obtain the partial derivatives $\frac{\delta f}{\delta x}$ and $\frac{\delta f}{\delta y}$. These two elements generate a two-sided ideal in $\mathbb{C}\langle x, y \rangle$, and we denote the closure of this ideal by $\langle \frac{\delta f}{\delta x}, \frac{\delta f}{\delta y} \rangle$. The resulting quotient

$$\text{Jac}(f) = \frac{\mathbb{C}\langle x, y \rangle}{\langle \frac{\delta f}{\delta x}, \frac{\delta f}{\delta y} \rangle}$$

is called the *Jacobi algebra* of f , and f is called the *potential*.

If the obtained Jacobi algebra is finite dimensional, then by the fundamental property of free algebras it can be presented as a homomorphic image of a finitely generated free algebra. By the First isomorphism theorem, it can then be presented as a factor algebra of this free algebra by the ideal generated by some relations (so we say that this algebra can be presented using generators and relations).

Recall also the definition of the closure of an ideal (see Section 2 in [5], page 7, formula (2.3)): Let I be an ideal in $\mathbb{C}\langle x, y \rangle$. Then the closure of I is the set of all elements $r \in \mathbb{C}\langle x, y \rangle$ such that for every n , $r \in I + F[n]$, where $F[n]$ is the linear space space spanned by all monomials from $\mathbb{C}\langle x, y \rangle$ whose degree is at least n .

Contraction algebras were introduced in [6] by M. Wemyss and W. Donovan, as invariants that can be associated to flops. It is known that every contraction algebra is a Jacobi algebra (see end of page 5 in [2]), so contraction algebras can be described by purely algebraic means, using generators and relations. Moreover, in [6] Donovan and Wemyss give several examples explaining how to find the potential of a given contraction algebra, and they give definition of the contraction algebra in Definition 2.9 in [6].

It was conjectured by Donovan and Wemyss and by Brown and Wemyss that all finite dimensional Jacobi algebras come from geometry as contraction algebras (see e.g., [2], page 8). There are some interesting results that suggest that this conjecture is true (see Theorem B in [4] and [2]).

In this section, the contraction algebra is obtained from the potential $f = \frac{1}{4}x^4 + x \cdot y^2$. It can be calculated that the Jacobi algebra $\text{Jac}(f)$ equals the quotient of the free $\mathbb{C}$ -algebra generated two elements x, y , by the ideal generated by relations: $x^3 + y^2, y^3, xy + yx$.

We now prove the conjecture given by M. Wemyss [13].

Theorem 4.2. Let

$$N = \frac{\mathbb{C}\langle x, y \rangle}{\langle xy + yx, x^3 + y^2, y^3 \rangle}.$$

Then N deforms flatly into $A = M_2(\mathbb{C}) \oplus \mathbb{C}^{\oplus 5}$.

We prove this theorem through a series of lemmas. Firstly, we fix notation in addition to the notation of Theorem 4.2.

Notation 4.3. Let i_1, i_2 be roots of the polynomial $x^2 - x + 1$.

Notation 4.4. Let $e \in \mathbb{C}$ be such that for $c = \frac{i_1 - i_2}{\sqrt{1 + e^2}}$ the polynomial

$$g(z) = (2z - 1)^2 z^3 + c^2$$

has 5 distinct roots $\alpha_1, \dots, \alpha_5 \in \mathbb{C}$. Further, we assume that $e \neq 0, e^2 \neq -1, c^2 \notin \{3, -1, -3\}$ and $\alpha_i \neq \frac{1}{2}$. Notice that $g(z)$ and $z^3 + 1$ have no common roots, so $\alpha_i \neq -1$ and $\alpha_j \notin \{i_1, i_2\}$.

Notation 4.5. Let $\alpha = (\alpha_1, \alpha_2, \alpha_3, \alpha_4, \alpha_5)$ and $\beta = (\beta_1, \beta_2, \beta_3, \beta_4, \beta_5)$ for $\beta_i \in \mathbb{C}$ such that

$$(\beta_1, \beta_2, \beta_3, \beta_4, \beta_5)(2\alpha_1 - 1, 2\alpha_2 - 1, 2\alpha_3 - 1, 2\alpha_4 - 1, 2\alpha_5 - 1) = c(1, 1, 1, 1, 1).$$

Notice that such β_i exist since $\alpha_i \neq \frac{1}{2}$.

Lemma 4.6. Let $\xi : \mathbb{C}\langle x, y \rangle[t] \rightarrow A$ be a homomorphism of $\mathbb{C}$ -algebras such that

$$\xi(x) = \begin{pmatrix} i_1 & 0 \\ 0 & i_2 \end{pmatrix} + \alpha, \quad \xi(y) = \frac{1}{\sqrt{1+e^2}} \begin{pmatrix} 1 & e \\ e & -1 \end{pmatrix} + \beta, \quad \xi(t) = \mathbb{1}.$$

Then $\xi(x)$, $\xi(y)$ and $\mathbb{1}$ generate A as an algebra.

Proof. Observe that $\xi(x^j(x^2 - x + 1)) = \alpha^j(\alpha^2 - \alpha + 1)$ (where $\alpha^j(\alpha^2 - \alpha + 1) = (\alpha_1^j(\alpha_1^2 - \alpha_1 + 1), \dots, \alpha_5^j(\alpha_5^2 - \alpha_5 + 1))$). Now let W be the matrix with entries $W_{ij} = \alpha_i^j(\alpha_i^2 - \alpha_i + 1)$. Let V be the Vandermonde matrix with entries $V_{ij} = \alpha_i^j$. Let D be a diagonal matrix with entries $D_{ij} = \delta_{ij}(\alpha_i^2 - \alpha_i + 1)$. It follows that $W = DV$. Now since V is nonsingular, W is nonsingular. Now this implies that any 5-dimensional vector in A is in $\text{im } \xi$. Then $\xi(x) - \alpha \in \text{im } \xi$ and hence

$$\begin{pmatrix} i_1 & 0 \\ 0 & i_2 \end{pmatrix} \in \text{im } \xi.$$

Similarly, $\xi(y) - \beta \in \text{im } \xi$ and hence

$$\begin{pmatrix} 1 & e \\ e & -1 \end{pmatrix} \in \text{im } \xi.$$

Now recall that i_1, i_2 are roots of the polynomial $z^3 + 1$, so we have

$$\begin{pmatrix} i_1 & 0 \\ 0 & i_2 \end{pmatrix}^3 = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Notice that every diagonal matrix is a linear combination of $\begin{pmatrix} i_1 & 0 \\ 0 & i_2 \end{pmatrix}$ and $\begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$. Now, it can be checked that these matrices are sufficient to generate all matrices in $M_2(\mathbb{C})$. ■

Notation 4.7. Let $f : \mathbb{C}\langle x, y \rangle[t] \rightarrow A[t]$ be a homomorphism of $\mathbb{C}$ -algebras such that

$$f(x) = t^2 \begin{pmatrix} i_1 & 0 \\ 0 & i_2 \end{pmatrix} + t^2 \alpha, \quad f(y) = \frac{1}{\sqrt{1+e^2}} t^3 \begin{pmatrix} 1 & e \\ e & -1 \end{pmatrix} + t^3 \beta, \quad f(t) = t \mathbb{1},$$

where $\mathbb{1}$ is the identity element in A . We write $\tilde{1} := (1, 1, 1, 1, 1)$. As usual we denote by 1 the identity element in $\mathbb{C}$ and thus in $\mathbb{C}\langle x, y \rangle[t]$.

We now prove Theorem 4.2.

Proof. Referring to Notation 4.5 and 4.7 we calculate:

$$\begin{aligned} f(x^2) &= t^4 \left(\begin{pmatrix} i_1^2 & 0 \\ 0 & i_2^2 \end{pmatrix} + \alpha^2 \right), \\ f(y^2) &= t^6 \left(\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + \beta^2 \right), \\ f(xy) &= \frac{t^5}{\sqrt{1+e^2}} \left(\begin{pmatrix} i_1 & i_1 e \\ i_2 e & -i_2 \end{pmatrix} + \alpha \beta \right), \\ f(yx) &= \frac{t^5}{\sqrt{1+e^2}} \left(\begin{pmatrix} i_1 & i_2 e \\ i_1 e & -i_2 \end{pmatrix} + \alpha \beta \right). \end{aligned}$$

Note that $f(x^2 - t^2x + t^4) = t^4(\alpha^2 - \alpha + \tilde{1})$ and hence

$$\begin{aligned} f(x(x^2 - t^2x + t^4)) &= t^6\alpha(\alpha^2 - \alpha + \tilde{1}), \\ f(x^2(x^2 - t^2x + t^4)) &= t^8\alpha^2(\alpha^2 - \alpha + \tilde{1}), \\ f(y(x^2 - t^2x + t^4)) &= t^7\beta(\alpha^2 - \alpha + \tilde{1}). \end{aligned}$$

Observe that

$$\begin{aligned} f(xy + yx) &= \frac{t^5}{\sqrt{1+e^2}} \begin{pmatrix} 2i_1 & e \\ e & -2i_2 \end{pmatrix} + t^5 2\alpha\beta, \\ f(t^2y) &= \frac{t^5}{\sqrt{1+e^2}} \begin{pmatrix} 1 & e \\ e & -1 \end{pmatrix} + t^5\beta, \\ f(t^5 \frac{i_2 - i_1}{\sqrt{1+e^2}}) &= \frac{t^5}{\sqrt{1+e^2}} \begin{pmatrix} i_2 - i_1 & 0 \\ 0 & i_2 - i_1 \end{pmatrix} + \frac{t^5(i_2 - i_1)}{\sqrt{1+e^2}} \tilde{1}. \end{aligned}$$

Note that $2i_1 - 1 + (i_2 - i_1) = 0$ and $-2i_2 + 1 + (i_2 - i_1) = 0$, so

$$f(xy + yx - t^2y + \frac{t^5(i_2 - i_1)}{\sqrt{1+e^2}}) = 2t^5\alpha\beta - t^5\beta + \frac{t^5(i_2 - i_1)}{\sqrt{1+e^2}} \tilde{1}.$$

Now notice that $2\alpha\beta - \beta + \frac{i_2 - i_1}{\sqrt{1+e^2}} \tilde{1} = 0$ by assumptions of Notation 4.5. It follows that $xy + yx - t^2y + \frac{t^5(i_2 - i_1)}{\sqrt{1+e^2}} \in I$ and so $xy + yx \in \langle I \rangle$ is a relation in N as defined in Notation 3.3.

Notice that by Notation 4.5 we have $\beta^2(2\alpha - 1)^2 = c^2\tilde{1}$. Now it follows by Notation 4.4 that

$$\alpha^3 + \beta^2 = 0, \quad (6)$$

and thus $x^3 + y^2 \in \langle I \rangle$ is a relation in N .

We now proceed to show that $y^3 \in \langle I \rangle$. Notice that:

$$f(y^3) = \frac{t^9}{\sqrt{1+e^2}} \begin{pmatrix} 1 & e \\ e & -1 \end{pmatrix} + t^9\beta^3.$$

Hence $y^3 - t^6y = t^9(\beta^3 - \beta)$. Therefore it is sufficient to show that $y^3 - t^6y + q \in I$ for some $q \in t\mathbb{C}\langle x, y \rangle[t]$. We denote:

$$\begin{aligned} r_1 &:= t^5(x^2 - t^2x + t^4) \\ r_2 &:= t^3x(x^2 - t^2x + t^4) \\ r_3 &:= tx^2(x^2 - t^2x + t^4) \\ r_4 &:= t^2y(x^2 - t^2x + t^4) \end{aligned}$$

Notice that the elements r_1, r_2, r_3 and r_4 are in $t\mathbb{C}\langle x, y \rangle[t]$. Note that it suffices to show that $y^3 - t^6y + q \in I$ for some $q \in \mathbb{C}r_1 + \mathbb{C}r_2 + \mathbb{C}r_3 + \mathbb{C}r_4$. Therefore it suffices to show that

$$\beta^3 - \beta \in \mathbb{C}e_1 + \mathbb{C}e_2 + \mathbb{C}e_3 + \mathbb{C}e_4 \quad (7)$$

where e_1, e_2, e_3, e_4 are such that

$$\begin{aligned} f(r_1) &= t^9(\alpha^2 - \alpha + \tilde{1}) =: t^9e_1, \\ f(r_2) &= t^9\alpha(\alpha^2 - \alpha + \tilde{1}) =: t^9e_2, \\ f(r_3) &= t^9\alpha^2(\alpha^2 - \alpha + \tilde{1}) =: t^9e_3, \\ f(r_4) &= t^9\beta(\alpha^2 - \alpha + \tilde{1}) =: t^9e_4. \end{aligned}$$

Now relation (7) is equivalent to

$$(2\alpha - 1)\beta(\beta^2 - 1) \in \mathbb{C}(2\alpha - 1)e_1 + \mathbb{C}(2\alpha - 1)e_2 + \mathbb{C}(2\alpha - 1)e_3 + \mathbb{C}(2\alpha - 1)e_4,$$

which in turn is equivalent to

$$-c(\alpha^3 + 1) \in \mathbb{C}(2\alpha - 1)e_1 + \mathbb{C}(2\alpha - 1)e_2 + \mathbb{C}(2\alpha - 1)e_3 + \mathbb{C}(2\alpha - 1)e_4. \quad (8)$$

Now, notice that $\alpha^3 + 1 = (\alpha + 1)(\alpha^2 - \alpha + 1)$ and that $\alpha + 1$ has non-zero entries, so multiplying relation (8) by $\alpha + 1$ we get an equivalent relation

$$\begin{aligned} (\alpha^3 + 1)(\alpha + 1) &\in \mathbb{C}(2\alpha - 1)(\alpha^3 + 1) + \mathbb{C}(2\alpha - 1)(\alpha^3 + 1)\alpha \\ &\quad + \mathbb{C}(2\alpha - 1)(\alpha^3 + 1)\alpha^2 + \mathbb{C}(\alpha^3 + 1). \end{aligned}$$

Notice that the right hand side can be rewritten as $\mathbb{C}(\alpha^3 + 1) + \mathbb{C}(\alpha^3 + 1)\alpha + \mathbb{C}(\alpha^3 + 1)\alpha^2 + \mathbb{C}(\alpha^3 + 1)\alpha^3$.

By Notation 3.3 we have $N = \mathbb{C}\langle x, y \rangle / \langle J \rangle$ and $J \subseteq \mathbb{C}\langle x, y \rangle$. Recall that, by Lemma 3.4, $\langle J \rangle \subseteq I + t\mathbb{C}\langle x, y \rangle[t]$, and $I = \ker f$ (for our function f). We have proved that $x^3 + y^2, xy + yx, y^3 \in \langle J \rangle$. Observe that the factor algebra $\mathbb{C}\langle x, y \rangle / \langle x^3 + y^2, xy + yx, y^3 \rangle$ has dimension 9. We also know that $N = \mathbb{C}\langle x, y \rangle / \langle J \rangle$ has dimension 9 (by Proposition 3.5 it has the same dimension as A), so $N = \mathbb{C}\langle x, y \rangle / \langle x^3 + y^2, xy + yx, y^3 \rangle$. Observe that if u is a product of elements α and β , then $t^j u \in \text{im} f$ for all sufficiently large j . This follows because, for every k ,

$$\alpha^k(\alpha^2 - \alpha + 1)t^{2k+4} = f(x^k(x^2 - t^2x + t^4)) \in \text{im} f,$$

and by the arguments regarding the Vandermonde matrix from the beginning of this proof.

Observe that if $r \in A$ is a product of any number of elements

$$B := \begin{pmatrix} i_1 & 0 \\ 0 & i_2 \end{pmatrix} \quad \text{and} \quad C := \begin{pmatrix} 1 & e \\ e & -1 \end{pmatrix}$$

then there is i such that $t^j r \in \text{im}(f)$ for all $j > i$ (and it is a product of elements $f(xt^{i_1})$ and $f(yt^{i_2})$ minus some product of elements αt^{i_3} and βt^{i_4} for some $i_1, i_2, i_3, i_4 \geq 0$). Since B and C generate the algebra A it follows that for every $r \in A$ there is i such that $rt^i \in \text{im}(f)$.

Let $R\langle\langle t \rangle\rangle$ denote a power series ring over an algebra R . By Theorem 3.7 there is an algebra R isomorphic to N and a deformation $R\langle\langle t \rangle\rangle$, which at $t = 0$ gives an algebra isomorphic to N . Moreover, by Theorem 3.10 this deformation at $t = 1$ gives algebra A . The deformation is flat by Remark 3.12. ■

5 A More General Method

In this section we consider a more general method that results in algebras N that are not necessarily local and have more generators. Furthermore, we consider A to have finitely many generators.

Theorem 5.1. Let A be a finitely generated $\mathbb{C}$ -algebra. Further, we assume and define the following:

1) Let F be a free $\mathbb{C}$ -algebra with j generators $x_1, \dots, x_j$ for some $j \in \mathbb{N}$. We define:

$$\begin{aligned} f : F[t] &\rightarrow A[t] \\ x_i &\mapsto \sum_{l=0}^m a_{i,l} t^l \end{aligned}$$

for some $m \in \mathbb{N}$ and some $a_{i,l} \in A$. We assume that for each $z \in \mathbb{C}$, $z \neq 0$ elements $\sum_{l=0}^m a_{i,l} z^l$ for $i = 1, 2, \dots, j$ generate A . In other words, images of the generators generate A when t is set to non-zero numbers.

2) We define

$$\mathcal{N} := \frac{\mathbb{C}\langle x_1, \dots, x_j \rangle[t]}{I},$$

where $I := \ker(f)$.

3) We assume that for every $r \in A$ there exists $i = i(r)$ such that $rt^i \in \text{im} f$.

Then our algebra $N = N_0$ is **isomorphic to**

$$\frac{\mathcal{N}}{\langle t + I \rangle}$$

and deforms flatly and associatively to A . Moreover N is isomorphic to $\frac{\mathbb{C}\langle x_1, \dots, x_j \rangle[t]}{\langle I, t \rangle}$, where $\langle I, t \rangle$ is the ideal of $\mathbb{C}\langle x_1, \dots, x_j \rangle[t]$ generated by elements of I and by t .

Proof. When reasoning as in the previous chapters we will use at all places $\mathbb{C}\langle x_1, \dots, x_j \rangle$ instead of $\mathbb{C}\langle x, y \rangle$, and we will use $\mathbb{C}\langle x_1, \dots, x_j \rangle[t]$ instead of $\mathbb{C}\langle x, y \rangle[t]$. Let A have dimension n over $\mathbb{C}$. Recall that we set

$$\mathcal{N} := \frac{\mathbb{C}\langle x_1, \dots, x_j \rangle[t]}{I},$$

where $I := \ker(f)$.

- Reasoning analogously as in Proposition 3.1 we obtain that there are $q_1, \dots, q_n \in F[t]$ such that

$$f(F[t]) \subseteq \sum_{i=1}^n \mathbb{C}[t]f(q_i),$$

and hence, since $I = \ker f$, we get

$$F[t] \subseteq \sum_{i=1}^n \mathbb{C}[t]q_i + I.$$

In particular, there are $\zeta_{i,k,m} \in \mathbb{C}$, $\xi_{i,k,m}(t) \in \mathbb{C}[t]$ such that

$$q_k \cdot q_m - \sum_{i=1}^n (\zeta_{i,k,m} q_i + t \xi_{i,k,m} q_i) \in I.$$

- Reasoning similarly as in Theorem 3.7 we obtain that if $d_1, \dots, d_n$ are free generators then

$$d_k * d_m - \sum_{i=1}^n (\zeta_{i,k,m} d_i + t \xi_{i,k,m} d_i),$$

gives a formal deformation such that $*_0$ gives a multiplication on N where N is the $\mathbb{C}$ -algebra with generators $d_1, \dots, d_n$ subject to relations

$$d_k *_0 d_m = \sum_{i=1}^n \zeta_{i,k,m} d_i.$$

Observe that $*$ defines a multiplication on the algebra $\mathbb{C}\langle d_1, \dots, d_n \rangle[t]/I'$, where I' is the ideal of $\mathbb{C}\langle d_1, \dots, d_n \rangle[t]$ generated by elements

$$d_k * d_m - \sum_{i=1}^n (\zeta_{i,k,m} d_i + t \xi_{i,k,m} d_i).$$

Notice that $d_1, \dots, d_n$ are free generators of the free algebra $\mathbb{C}\langle d_1, \dots, d_n \rangle$. Notice that the algebra $\mathbb{C}\langle d_1, \dots, d_n \rangle[t]/I'$ is isomorphic to algebra $\mathcal{N}$. To see this consider the map $\sigma : \mathbb{C}\langle d_1, \dots, d_n \rangle[t] \rightarrow \mathcal{N}$

given by

$$d_i \rightarrow q_i + I.$$

Notice that $I' \subseteq \ker \sigma$ since

$$(q_k + I) * (q_m + I) = \sum_{i=1}^n \zeta_{i,k,m}(q_i + I) + t \xi_{i,k,m}(q_i + I).$$

Recall that $\ker \sigma = I'$ because if $a \in \ker \sigma$ then by using relations from I' we can present a as $a' + i$ where $i' \in I$ and where $a' = \sum_{i=1}^n \alpha_i(t) d_i$ for some $\alpha_i(t) \in \mathbb{C}[t]$. Notice that then $\sigma(i') = 0$ and hence

$$\sigma(a) = \sigma(a') = \sigma\left(\sum_{i=1}^n \alpha_i(t) d_i\right) = \sum_{i=1}^n \alpha_i(t) q_i.$$

Reasoning as in Corollary 3.3 we get that this implies that all $\alpha_i = 0$, hence $a' = 0$, hence $a = i'$. Therefore, $\ker \sigma \subseteq I$ and consequently $\ker \sigma = I'$.

Therefore, by the First Isomorphism Theorem for algebras $\mathbb{C}\langle d_1, \dots, d_n \rangle[t]/I'$ is isomorphic to $\text{Im}(\sigma)$, which is equal to $\mathcal{N}$. To see that σ is surjective, notice that $q_1 + I, \dots, q_n + I$ span $\mathcal{N}$ as a linear space (as mentioned before this can be proved similarly as Proposition 3.1).

- Reasoning as in Proposition 3.5 we obtain that N is n dimensional $\mathbb{C}$ -algebra.
- Reasoning similarly as in Proposition 3.6 we obtain that the algebra N is isomorphic to the algebra $\frac{\mathbb{C}\langle x_1, \dots, x_j \rangle[t]}{(I, t)}$ where (I, t) is the ideal of $\mathbb{C}\langle x_1, \dots, x_j \rangle[t]$ generated by elements from I and by t . By the Third Isomorphism Theorem N is isomorphic to

$$\frac{\mathcal{N}}{(t + I)}.$$

This can be seen by reasoning similarly as in Proposition 3.9 but taking t instead of $t - 1$ in the proof.

- We will now consider deformations $*_1$ at $t = 1$. Consider first the homomorphism of $\mathbb{C}$ -algebras $\xi : F[t] \rightarrow A$ given by $\xi(x_i) = \sum_{l=0}^m a_{i,l}$ for $i = 1, 2, \dots, j$. Reasoning similarly as in Proposition 3.8 we obtain that $\ker \xi = (I, t - 1)$. Reasoning analogously as in the last two lines of Proposition 3.9 we obtain that $F[t]/\ker \xi$ is isomorphic to A .
- Reasoning as in Theorem 3.10 we see that the algebra $\mathbb{C}\langle d_1, \dots, d_n \rangle[t]/I$ deforms at $t = 1$ to the algebra $F[t]/(I, t - 1) = F[t]/\ker \xi$, which is isomorphic to A .
- We will now consider deformations $*_z$ at $t = z$ where $z \in \mathbb{C}, z \neq 0, z \neq 1$. Consider first the homomorphism of $\mathbb{C}$ -algebras $\xi : F[t] \rightarrow A$ given by $\xi(x_i) = \sum_{l=0}^m a_{i,l} z^l$ for $i = 1, 2, \dots, j$. Reasoning similarly as in Proposition 3.8 we obtain that $\ker \xi = (I, t - z)$ (this can be done by using $t' = t/z$ at the place of t in the proof of Proposition 3.8, and in the first line declaring that $p_i \in F$ instead of $p_i \in \mathbb{C}\langle x, y \rangle$).

Next, reasoning analogously as in the last two lines of Proposition 3.9 we obtain that $F[t]/\ker \xi$ is isomorphic to A (this follows since elements $\xi(x_i)$ generate A).

Next, reasoning as in Theorem 3.10 we see that the algebra $\mathbb{C}\langle d_1, \dots, d_n \rangle[t]/I$ deforms at $t = z$ to the algebra $F[t]/(I, t - z) = F[t]/\ker \xi$, which is isomorphic to A . ■

Note that if $j = 2$ then the method described above corresponds to the method described in section 3 provided that $a_{i,0} = 0$ for all i .

6 Future Work

In this section we pose some questions about Method 1 and suggest some generalisations.

Question 6.1. Let A be a $\mathbb{C}$ -algebra and suppose $G_1 = \{a, b\}$ and $G_2 = \{a', b'\}$ are two distinct generating sets of A . Under what conditions does Method 1 using A, G_1 and A, G_2 result in the same algebra N_0 that has A as a deformation?

Question 6.2. Let A be a $\mathbb{C}$ -algebra. Under what conditions does Method 1 only produce one algebra N that has A as a deformation?

Question 6.3. Let A be a $\mathbb{C}$ -algebra. Does there exist a finitely terminating algorithm that will produce all the algebras N arising from Method 1 that have A as a deformation?

Question 6.4. Let A be a $\mathbb{C}$ -algebra. Do ring theoretic properties of the generators a, b (for example being idempotent, being irreducible idempotent, prime) determine any properties of the algebra N arising from Method 1 using $A, \{a, b\}$?

Question 6.5. Let A be a $\mathbb{C}$ -algebra. Do ring theoretic properties of A (for example being semi-simple, local) determine any properties of the algebra N arising from Method 1 using A ?

Question 6.6. Can Method 1 be generalised to be used for algebras over an arbitrary field (or over an arbitrary commutative ring)?

Funding

The third author acknowledges support from the EPSRC programme grant [EP/R034826/1]; and from the EPSRC research grant [EP/V008129/1].

Acknowledgments

We are grateful to Michael Wemyss for suggesting his interesting questions, which inspired this paper, and for useful comments about contraction algebras and their applications in geometry. We would also like to sincerely thank the referee, whose comments greatly improved this paper.

References

1. Bergman, G. M. "The diamond lemma for ring theory." *Adv. Math.* **29**, no. 2 (1978): 178–218. [https://doi.org/10.1016/0001-8708\(78\)90010-5](https://doi.org/10.1016/0001-8708(78)90010-5).
2. Brown, G. and M. Wemyss. "Local normal forms of noncommutative functions." (2022): preprint, arXiv:2111.05900.
3. Coll, V., M. Gerstenhaber, and A. Giaquinto. "An explicit deformation formula with noncommuting derivations." *Israel Math. Conf. Proc.* **1** (1989): 396–403.
4. Davison, B. "Refined invariants of finite-dimensional Jacobi algebras." (2023): preprint, arXiv:1903.00659.
5. Derksen, H., J. Weyman, and A. Zelevinsky. "Quivers with potentials and their representations I: mutations." *Selecta Math. (N.S.)* **14**, no. 1 (2008): 59–119. <https://doi.org/10.1007/s00029-008-0057-9>.
6. Donovan, W. and M. Wemyss. "Noncommutative deformations and flops." *Duke Math. J.* **165**, no. 8 (2013): arXiv:1309.0698. <https://doi.org/10.1215/00127094-3449887>.
7. Dotsenko, P. T. V. "Tangent complexes and the diamond lemma." (2020): preprint, arXiv:2010.14792.
8. Feigin, B. and A. Odesski. "Flat deformations of algebras and functional equations." *J. Comb. Algebra* **3**, no. 3 (2019): 215–36. <https://doi.org/10.4171/jca/31>.
9. Hua, Z. and Y. Toda. "Contraction algebra and invariants of singularities." *Int. Math. Res. Not. IMRN* **2018** (2018): 3173–98. <https://doi.org/10.1093/imrn/rnw333>.
10. Nabijou, N. and M. Wemyss. "GV and GW invariants via the enhanced movable cone." (2023): preprint, arXiv:2109.13289.
11. Shepler, A. V. and S. Witherspoon. "PBW deformations of skew group algebras in positive characteristic." *Algebras Representation Theory* **18**, no. 1 (2015): 257–80. <https://doi.org/10.1007/s10468-014-9492-9>.
12. Toda, Y. "Non-commutative width and Gopakumar–Vafa invariants." *Manuscripta Math.* **148**, no. 3–4 (2015): 521–33. <https://doi.org/10.1007/s00229-015-0760-8>.
13. Wemyss, M. Personal communication, 2023.
14. Wemyss, M. "A lockdown survey on cDV singularities." (2023): preprint, arXiv:2103.16990.
15. Witherspoon, S. J. *Hochschild Cohomology for Algebras*. Graduate Studies in Mathematics, vol. 204. Providence, RI: American Mathematical Society, 2019.